URBANVERSE

MEMBER 2 | FRONTEND + 3D DIGITAL TWIN

Project: City-Scale Digital Twin for Multi-Hazard Prediction and Urban Resilience Planning

Hackathon target: Build a convincing, working flood-risk MVP for a small pilot area in Mangalore. Use real map/geographical data where practical and clearly label simulated rainfall and risk outputs.

Your mission

Build the part judges see: a usable dashboard, map, rainfall control and clear risk results.

Your tasks

- Create the React dashboard inside frontend/.
- Display the selected Mangalore pilot area on a map.
- Add a rainfall slider or input measured in millimetres.
- Add a Run Simulation button that calls POST /api/simulate.
- Display risk score, risk level, affected zones and recommendations.
- Show zone risk on the map using colours or a heatmap layer.
- Add a 3D view only after the 2D map and simulation flow work.

What you must hand over

- A frontend that starts locally.
- A map with the pilot area visible.
- A rainfall control and Run Simulation button.
- A results panel and risk-coloured zones.
- A clear loading/error state if the backend is unavailable.

File ownership and coordination

- Own frontend/.
- Use the API contract; do not hardcode a different response format.
- Use mock JSON temporarily if the backend is not ready, but keep the same field names.
- Coordinate with Member 3 before relying on GeoJSON property names.

Start in this order

1. Create the dashboard layout and map first.
2. Use sample GeoJSON and mock simulation JSON to build the UI.
3. Send Member 1 the exact API request you will make.
4. Connect the real API as soon as /api/health and /api/simulate are available.
5. Add 3D only if the core flow is already working.

Your acceptance test

Set rainfall to two different values, run both simulations, and confirm the displayed scores/colours update from the API response rather than from hardcoded UI values.

Team rule

Commit working changes to your own Git branch, push them, and tell the integration lead what changed. Do not edit another member's files without agreement. If you are blocked, share the exact error and the smallest reproducible example rather than silently changing the shared interface.